

Combining Spatial and Non-Spatial Sonification for Sound Localization in Virtual Environment: a Subjective Study

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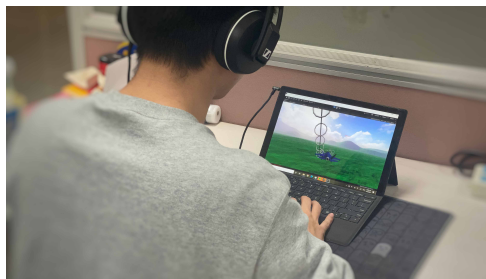
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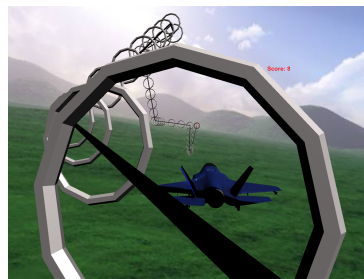
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(a)



(b)



(c)

Figure 1: (a) Experiment setup. The participant sits in front of a laptop with stereo headphone and interact with it using keyboard when secondary task is introduced (study II and III). (b) Secondary task scene for study II. The participant is asked to maneuver the aeroplane to pass the rings stick to the target aviation path as close as possible during the experiment. (c) Secondary task scene for study III. The participant is asked to drive the car on the highway, avoiding other vehicle on the road.

ABSTRACT

Spatially sonified audio is frequently used in virtual environments to serve as an object localization cue. However, spatializers that are available to the general public typically have their limitations and lead to sound localization errors. By combining it with non-spatial sonification, the technique of encoding information with perceptual attributes of the sound, the localization errors of spatially sonified audio could be reduced. However, the encoded information would

require higher mental effort to understand, which could be a concern under high task load. In this paper, we conducted a subjective study to compare spatial sonification against combined auditory cue using spatial and non-spatial sonification for target localization. We tested both methods under low and high task load. Our results show that combined sonification could significantly reduce the errors in localization, but the advantage would be influenced by higher task load. Further, we compared several different non-spatial sonification methods for encoding depth information. Our experiment shows that both pitch and pulse frequency modulation are feasible for presenting depth of the target. Our results could contribute as a first step to a general guideline of designing sonification method as target localization cue.

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CHI '22 Extended Abstracts, April 29-May 5, 2022, New Orleans, LA, USA

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ACM ISBN 978-1-4503-9156-6/22/04.

<https://doi.org/10.1145/3491101.3519853>

CCS CONCEPTS

• **Human-centered computing** → **Usability testing**; **Auditory feedback**; **Virtual reality**.

KEYWORDS

sound localization, spatial audio, virtual surround sound, sonification, subjective quality assessment, user experience

ACM Reference Format:

Qi Song, Naiming Yao, Zihao Feng, Hui Chen, and Hongan Wang. 2022. Combining Spatial and Non-Spatial Sonification for Sound Localization in Virtual Environment: a Subjective Study. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts (CHI '22 Extended Abstracts)*, April 29-May 5, 2022, New Orleans, LA, USA. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3491101.3519853>

1 INTRODUCTION

Auditory cues play many important roles for interaction in virtual environment, one of which is providing the location of the object [14]. Physical surround sound are less frequently used in practical due to its hardware limitation; instead, simulated spatial audio is frequently used in virtual environment given the fact that stereo headphones are widely available to the user. Such simulated surround sound is mostly known as virtual surround sound or spatial audio. It utilizes sonification technique to manipulate intensity and phase difference of the sound mimicking localization cues of natural spatial hearing. Therefore, it is also called as spatial sonification, distinguished against its non-spatial sonification counterpart [21]. Though spatially sonified audio is widely used in commercial products, it is also known to have more localization errors in the 'cone of confusion' [14, 26]. Such cone of confusion mostly refers to three types of confusions: front-back, up-below and depth. Figure 2(a) shows the cone of confusion studied in study I and II, in which front-back and up-below confusion were focused.

Sonification is mostly recognized as a technique using non-speech audio to encode information and perceptualize data [12], such as audio interface for presenting data [3, 19], or auditory guidance systems [27]. When investigating sonification techniques as localization cues, Parseihian et al. [21] classified the techniques as spatial sonification and non-spatial sonification paradigm. Spatial sonification focuses on manipulating audio features to mimic natural spatial hearing; non-spatial sonification uses perceptual characteristics of the sound to encode guidance information. In their study, Parseihian et al. also noted that, non-spatial sonification generally requires longer learning process. Therefore, we investigated the usability of combined sonification compared to spatial sonification in study I, and further tested them under high task load in study II.

Further, as there is no natural link between the sonification method and the encoded information, metaphoric relationship is vital for designing a usable non-spatial sonification guidance method. However, Parseihian and colleagues' evaluation was conducted on a 1D guidance task on a tablet. For localizing information in 3 dimensional space, which method is the best metaphor of each dimension stays unclear. Starting from depth, which is one of a dimension suffers from localization errors of spatial sonification, we evaluated several different sonification methods to find out the most feasible sonification analogy to depth information in study III.

Through the results of our user studies, we present the following contribution and provide insights into the performance of combined sonification for target localization under different levels of task load.

Existing work mostly focuses on auditory guidance using relative feedback [21]; our study is gravitated to absolute localization cue. Without continuous interaction, achieving high localization performance is challenging for absolute sonification cues due to the lack of relative reference point. However, it is well suited for the cases in which the user does not need to further interact with the target.

- We compared the localization performance for spatial sonification cue with auditory cue combining non-spatial and spatial sonification. We verified that spatial sonification cue has localization ambiguity in the confusion cone, see figure 2(a). We showed that by combining non-spatial sonification cue with spatial sonification, localization performance in the cone of confusion was significantly improved.
- We showed that under higher task load (by introducing secondary task), no significant difference in localization performance between spatial sonification and combined cue was found. The influence on target localization performance from secondary task was significant. The advantage of combined cue was neutralized by the task load introduced by secondary task. The primary reason of such phenomenon could be that user does not have spare mental capacity under high task load to decode the information in non-spatial sonification.
- We investigated different sonification methods on depth dimension in a 3D localization task under high task load. We conducted experiment to compare which sonification method has the most feasible metaphorical connection to depth. We found significant shorter task completion time for both pitch modulation and pulse frequency modulation method compared to spatial sonification (loudness). The same approach could be applied to other sonification methods repeatedly until a feasible metaphoric sonification for each dimension is found and a combination is formed.

2 RELATED WORK

Our studies touch upon several field of research: spatial sonification guidance, non-spatial sonification guidance cue, which we describe below.

2.1 Spatial Sonification Guidance

The spatial sonification is based on human perceptual and cognitive ability for natural spatial hearing. Rayleigh [23] first proposed two interaural cues for binaural sound localization, namely interaural time difference (ITD) and interaural intensity difference (IID). ITD is produced when the sound travels longer to the rear ear, and IID is a result of shadowing effect cast by the head. IID and ITD are dominant factors for localizing sound source along the longitudinal axis. Our outer ears impose further spectral coloration on the incoming sound waves when it reaches the eardrum. Such coloration is proven to be vital for localizing sound source in the vertical plane and solving the front-back ambiguity [26]. The spatial sonification paradigm mimics spatial audio in reality by applying Head Related Transfer Function (HRTF) to the origin sound source. HRTFs are systematically measured transfer functions of sound source to eardrum in anechoic chamber [2, 25].

Spatial sonification is widely used in guidance systems. Previous work includes using spatial sonified auditory cue to support navigation when visual information is not available. For example, for blind people [15] or on wearable devices whose visual display is small thus not feasible for displaying navigation information [30]. Binetti et al. [4] adopted spatial sonification cue to mitigate cue-target depth disparity issue in augmented reality guidance systems. As spatial sonification utilizes our natural spatial hearing capabilities, guidance system using such paradigm could potentially have shallower learning curve and lower mental effort requirement. However, spatial sonification suffers from low resolution issue [21], especially for objects located near the confusion zone. This degraded performance is primarily a result of the usage of generic HRTF. As noted by Wightman and Kistler [29], ITD alone is not enough for localizing sound source unambiguously; the individual-dependent intensity attenuation recorded in HRTF is vital to localize the sound source. Subsequent work also [7, 18, 28] proved that spatial sonification cue using customized HRTF significantly outperforms the ones using generic HRTF sound localization performance. However, obtaining HRTF requires rigorous and systematic measurement with professional equipment, which means customized HRTF is not available to the general public. Commercial off-the-shelf spatializers [9, 10, 16, 22] typically use generic HRTF. As a result, using available spatial sonification tools would lead to localization ambiguities.

2.2 Non-spatial Sonification Guidance

Manipulating perceptual characteristics of the audio, to encode guidance information into the sonified audio, is recognized as non-spatial sonification paradigm. Parseihian and colleagues' work [21] provides a review of several frequently used non-spatial sonification strategies. In their study, 9 different non-spatial sonification methods has been compared and evaluated on a 1D guidance task. They have also noted that as there is no direct relation between the encoded information and the manipulated perceptual characteristics, metaphoric relationship is vital to establish high performance and easy to use non-spatial sonification strategy. Non-spatial sonification is widely employed for auditory interface and guidance. Wegner [27] used amplitude modulation to guide the surgeon, they also proposed the Wave Terrian, a sonification of the anatomical surface. Scholz et al. [24] modulated pitch and brightness to construct a 7x7 guidance matrix for stroke rehabilitation. Using pulse frequency to indicate distance where the time interval between two impulse tones increases with the distance is a widely used sonification strategy that could be found in most modern car parking aids [21].

Non-spatial sonification as a localization cue is also used in combination with other localization cues. By combining spatial sonification on longitudinal axis and pitch modulation for elevation axis, Black and colleagues have established an auditory guidance system for needle placement for surgical purpose [5]. In a more recent work in non-visual guidance method on augmented reality head-mounted display [17] different combination of non-spatial sonification cues and modulated vibrotactile cues have been investigated. Their result shows that presenting longitudinal and depth information using vibrotactile modality and providing latitudinal

information by modulating pitch and loudness of the audio is the best performing one.

3 SYSTEM APPROACH AND IMPLEMENTATION

In this work we tested auditory cue combining spatial and non-spatial sonification as a information localization cue under different levels of task load. In contrast to the relative feedback studied in the previous work [13, 21], information localization focuses on the absolute feedback. Understanding information localization performance with absolute sonification feedback would be useful for cases in which the user should be noticed with the presence and location of such object but does not required to further interact with it. This would enable numerous interactions in virtual environment and augmented reality, e.g. advanced driver assistant system that alerts the driver with nearby pedestrians.

We used Unity, version 2019.4.25 LTS ¹ together with DearVR engine 1.6 [22] to implement the system. DearVR is evaluated to be the best performing commercial off-the-shelf spatializer for localization performance [14], we employed it to implement spatial sonification for our experiment. For secondary task, in study II, we implemented a path following aviation game with Unity; in study III, we employed a simulated driving scene from Highway Racer [8]. The software runs on a ordinary laptop with stereo headphone for audio output.

3.1 Combined Sonification Method in Study I and II

In study I and II, we combined the spatial and non-spatial sonification cue (here and after referred as COMBINED) for 3D localization task. Particularly, the non-spatial sonification in the COMBINED method was introduced to mitigate the front-back confusion and up-below ambiguity of spatial sonification. Figure 2(b) provides a demonstration of the design of the COMBINED method. The spatial sonification serves as a foundation to COMBINED method, providing longitudinal information.

To mitigate front-back confusion, we used auditory brightness modulation. If the source is located in front of the user, the auditory cue would be brighter and vice versa. Auditory brightness is the auditory metaphor of visual brightness; brighter sound is achieved by having more energy distributed on the higher frequency bands for a same harmonic sound. We chose brightness because it is an analogy to visual brightness. Brighter auditory sound in the front could be linked to the illuminated display in front of the user, building stronger metaphoric connection between localization information and the non-spatial sonification cue. In doing so, common problems for non-spatial sonification could be mitigated, e.g. steep learning curve and requirement of higher mental effort to decode the information.

For up-below information, we used pitch modulation. We generally followed the design of latitudinal feedback [17], with two modification. We first removed the volume modulation and only employed the pitch modulation. We then adjusted the sonification method in the following manner. Marquardt and colleagues'

¹www.unity.com

work focused on interactive guidance. In their design, the pitch and volume reach the highest level when user's head direction points directly to the target. In our implementation, the pitch is highest when the sound source is above the user and lowest when the target is below the user.

3.2 Sonification Methods in Study III

In study III, we compared different sonification methods for depth information for localizing target on ground level. The longitudinal information is provided using spatial sonification. To encode depth information, we implemented three sonification strategy. For non-spatial sonification, we had pulse frequency and pitch modulation. As shown in figure 3(b)(A), pulse frequency refers to the frequency of alternating on and off pattern of an audio. When the object gets closer to the target, such alternation is accelerated and pulse frequency is increased. Such sonification strategy is frequently used in car parking aid [21], thus using pulse frequency to sonify depth information could result in shallower learning curve. We also had pitch modulation as one of the sonification strategy for depth information (figure 3(b)(B)). Closer object would have auditory cue with higher pitch and rear object with lower pitch. Third, we also included spatial sonification for comparison. In this case refers to the loudness (see figure 3(b)(C)). In the following context, we denote to the combination of each of the sonification method for depth with spatial sonification for longitudinal information as SPATIAL+PULSE, SPATIAL+PITCH, SPATIAL+LOUDNESS.

4 USER STUDY

We first conducted study I to validate that the COMBINED method could be used to mitigate localization errors in confusion zone caused by spatial sonification. Before each study begins, each participant was informed about the process and goal of the study, and signed an informed consent form. The experiment setup are shown in the figure 1(a). Post-experiment questionnaires include standard questionnaires like NASA-TLX [11] and System Usability Scale [6]. Study III also adopted customized questionnaires using 5-point Likert scale. For each study, learning effect was mitigated with random ordering. Study I and II evaluated sonification cue for latitudinal and longitudinal information, study II extended study I with secondary task. In study III, different modes were tested that encode information on ground level with different depth under high task load introduced with secondary task. Participants were instructed to speak aloud the perceived direction of the sound source. Accuracy measured as the hit rate of perceived direction with the actual direction. Task completion time was also recorded in study III.

4.1 Study I - Localization Performance under Low Task Load

RQ1: how is COMBINED sonification method performing for object localization task?

In this study, we compared COMBINED method with spatial sonification (here and after referred as SPATIAL). Only longitudinal and latitudinal directions were considered. 7 participants (4 male, 3 female) aged from 22 to 35 participated in the study recruited using volunteer recruit poster. No compensation was offered to the participants. Participants were asked to sit still and stare at a

cursor in the center of the monitor. They were told to speak aloud the direction of the sound source as soon as possible, but there was no time limit. One factorial with-subject experiment with random ordering was employed. Objects were placed in 8 longitudinal directions, plus above and below, see figure 2(a). Before each block, an introduction of the current sonification strategy together with a repeatable training section was provided. In each block, 5 trials were tested. The location of the target was generated randomly, but one direction appears at most once in a single block; in each block, at least 3 trials were from the confusion cone. In doing so, we could better understand the contribution of non-spatial sonification in COMBINED case, as they were introduced to mitigate front-back and up-below ambiguity experienced in spatial sonification. After each block, the participant was asked to fill NASA-TLX and System Usability Score.

4.2 Study II - Localization Performance under High Task Load

RQ2: how is COMBINED sonification method performing under high task load?

The same participants from study I took part in study II. In study II, user was asked to perform the same localization task under dual task setup. The user had to maneuver an aeroplane to follow a given path and performing localization using given sonified auditory cue. Such setup mimics everyday gaming experience where the player controls an avatar moving in 3D virtual space while paying partial attention to surrounding objects like enemies or treasures. Before study II, the user had a training section for maneuvering the aeroplane and could repeat this training section until he/she was satisfied. During the experiment, the localization task setup was the same as in study I study; the target path for maneuvering the aeroplane was the same for both blocks and the training section. For secondary task performance, mean deviation to the target path was calculated. After each block, the participant was asked to fill NASA-TLX and System Usability Score and was specifically asked to focus the task load and usability of the localization task.

4.3 Study III - Depth Localization under High Task Load

RQ3: Which sonification method is best performing while showing strong metaphoric link to depth information?

21 university students (9 male, 12 female) were recruited with recruiting poster for study III. In this study, we compared different sonification methods for encoding the depth of the target. We choose depth because the spatial sonification of depth information, loudness, would not only result in degraded localization performance on depth perception but also could have more negative impact. For rear object, low volume of the sonified auditory cue could prevent the user from hearing it clearly, rendering localization information on all of the dimensions inaccessible to the user. For near object, high loudness could overwhelm the user and cause hearing damage [20]. Study II employed one factorial within subject design. Three sonification mode, SPATIAL+PULSE, SPATIAL+PITCH, SPATIAL+LOUDNESS, were tested. For secondary task, the participant was asked to play a car racing game that mimics highway driving scene.

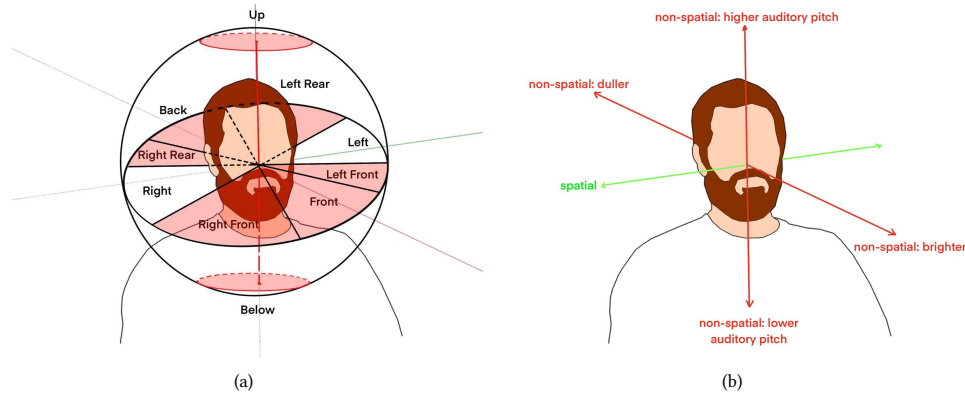


Figure 2: Demonstration of localization task setup in study I and II. (a) The possible locations of the target. Directions with non-zero projection on up-below and front-back axes are categorized as confusion cone of spatial sonification, and colored red (Front, Back, Left Front, Right Front, Left Rear, Right Rear, Up, Below); left and right are directions out of the cone of confusion. The cone of confusion are directions or areas on which most localization ambiguities are discovered for spatial sonification with generic HRTF [14, 26]. In study I and II, we focused on the up-below (elevation) and front-back ambiguity. (b) The COMBINED sonification cue employed in study I and II. Dimensions using non-sonification methods are colored red (vertical and front-back). On the vertical axis, auditory pitch modulation is used. For front-back information, we use auditory brightness modulation. For left and right, we use spatial sonification method.

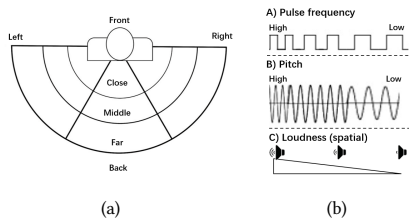


Figure 3: Localization task setup in study III. (a) Possible location of the target. The target is always located on the ground level and could be located at left, middle or right on the back side of the user, with three different levels of depth. (b) Three sonification methods of depth information used in study III. (A) Pulse frequency; (B) Pitch; (C) loudness.

5 RESULTS AND DISCUSSION

For study I, t-test was used to analyze the effect of sonification strategy on localization performance (measured with hit rate). For targets located in the confusion cone of spatial sonification, average hit rate of using COMBINED strategy (0.58) is significantly higher than SPATIAL strategy (0.14, p -value < 0.05). Thus, combining non-spatial sonification with spatial sonification could compensate the localization ambiguity in confusion cone. See left two columns of table 1 for further detail.

In study II, we extended the experiment conducted in study I with high task load case. Two-way repeated Measure ANOVA was used to measure the effect of sonification strategy and the secondary task on localization performance. Localization performance showed no significant difference on different sonification strategy used ($p > 0.05$). However, the secondary task showed significant negative

influence on the task performance ($p < 0.05$) when we compare the result of study II (with secondary task) to study I (without secondary task). Thus, answering RQ2 COMBINED and SPATIAL have similar information localization performance, COMBINED sonification method does not show feasibility under high work load. Other details of study II is reported in the right two columns of table 1.

Subjective feedback from participants in study I and II showed favor to COMBINED sonification, as subjects responded that SPATIAL requires more effort to distinguish target in the confusion cone. However, one of the participants pointed out that for COMBINED cue, effort is needed to connect the non-spatial sonification audio with the localization information. In addition, this participant advised that providing extra training section for the COMBINED method could further promote its performance. Another participant reported that he/she had previous gaming experiences with spatially sonified audio, the overlaid non-spatial sonification in the COMBINED cue disturbed him/her and resulted it degraded performance.

The degraded performance for COMBINED method induced by secondary task shows that decoding information from non-spatial sonification cue requires mental efforts, and higher work load would prevent the user from successfully processing it. Subsequent research on non-spatial sonification methods could pay more attention to metaphoric relationship between the sonification cue and the encoded localization information. Design of end system should also consider users with different experience and background: as discussed above, participants who are familiar with spatial sonification could be confused by the non-spatial sonification elements in COMBINED method.

For study III, both the combined methods (SPATIAL+PULSE and SPATIAL+PITCH) significantly outperformed the spatial sonification approach (SPATIAL+LOUDNESS) in terms of reaction time

	Without Secondary Task		With Secondary Task	
	COMBINED	SPATIAL	COMBINED	SPATIAL
Overall Localization Accuracy	0.71	0.43	0.51 ⁺	0.34
Accuracy in Confusion Cone	0.58*	0.14	0.32	0.29
Accuracy Out of Confusion Cone	0.86	0.60	0.81	0.52
Mean Path Deviation	N/A	N/A	1.66	1.84
NASA-TLX score	40.1	42.6	40.6	41.5
SUS score	65.0	62.5	63.2	63.9

Table 1: Result of study I (left two columns) and study II (right two columns), localization of object with COMBINED and SPATIAL cue without and with secondary task. ⁺ Secondary task significantly reduced overall localization accuracy. * Without secondary task, COMBINED method significantly outperforms SPATIAL localization accuracy for objects in the confusion cone.

	SPATIAL + PULSE	SPATIAL + PITCH	SPATIAL + LOUDNESS
Localization Accuracy	0.75	0.72	0.71
Reaction Time	5.40**	5.44**	6.58

Table 2: Localization performance in study III (p<0.001).**

($p < 0.001$, RM-ANOVA), see table 2. Mann-Whitney U test showed no significant effect of sonification method on sound localization accuracy. In subject evaluation, 38.1% of the user show preference to each of the combined sonification strategy. For depth perception, 38.1% of the total user prefer SPATIAL+PULSE and 42.68% of the user think SPATIAL+PITCH makes depth localization easier. Most users think there is no learning difficulties among the 3 sonification modes. In the earlier context we have discussed that the spatial sonification on depth dimension, loudness, has its drawback, which matches with observations from study II. In further subject interview, one of the participants reports that it was hard to obtain precise depth information from loudness modulation, resulting in longer reaction time. Concluding study II, both pitch and pulse frequency are promising non-spatial sonification method for encoding depth information.

6 LIMITATION AND FUTURE WORKS

In this study we investigated combining non-spatial and spatial sonification for target localization. Due to time limit, we only investigated a limited number of sonification methods. In the future work, we wish to systematically evaluate more combinations of sonification methods and most importantly to obtain a better understanding on how to establish a strong metaphorical link between the sonification and the encoded location information. On the other hand, we only tested the localization performance under limited number of levels of task difficulty. We wish to evaluate the system under various levels of difficulty and quantify the task load to understand the robustness of different sonification methods. Further, Baddeley [1] has noted that the working memory span consist of a "visual-spatial sketchpad" and a "phonological loop" under the higher level "central executive" component. In our study, the

introduced secondary task in study II and study III were both visual and spatial related, which would majorly occupy visual-spatial sketchpad. In subsequent work, it might be feasible to evaluate non-spatial sonification for localization under different types of secondary tasks to understand when and where the non-visual auditory cue is decoded to spatial localization information.

ACKNOWLEDGMENTS

This work was supported by the National Key R&D Program of China (2020YFC2004100)

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